

Exponential mixing of circle expansion

①

Let $f_M: \mathbb{T} \rightarrow \mathbb{T}$, $x \mapsto Mx$ ($M \geq 2$)

φ, ψ C^1 function on $\mathbb{T}$, then

$$\int \varphi \circ f_M^n \psi - \int \varphi \int \psi = O(|\varphi|_{C^1} |\psi|_{C^1} M^{-n}).$$

pf. Use Fourier expansion, $e_k(x) = e^{2\pi i k x}$ $k \in \mathbb{Z}$

$$\varphi = \sum \varphi_k e_k, \quad \psi = \sum \psi_\ell e_\ell.$$

Due to $\varphi, \psi \in C^1$

$$\varphi_k = \int \varphi(x) e^{-2\pi i k x} dx = \int \varphi(x) \frac{d}{dx} \frac{e^{-2\pi i k x}}{-2\pi i k} dx$$

$$= \frac{1}{2\pi i k} \int \varphi'(x) e^{-2\pi i k x} dx.$$

$$|\varphi_k| \leq \frac{1}{k} |\varphi'|_{C^1} \leq \frac{1}{k} |\varphi|_{C^1}.$$

Therefore.

$$\int \varphi \circ f_M^n \psi - \int \varphi \int \psi = \int \sum_k \varphi_k e_k \circ f_M^n \sum_\ell \psi_\ell e_\ell - \varphi_0 \psi_0$$

$$= \sum_{\substack{k, \ell \neq 0 \\ (k, \ell) \neq 0}} \varphi_k \psi_\ell \int e_k \circ f_M^n e_\ell$$

$$= \sum_{\substack{k, \ell \neq 0 \\ (k, \ell) \neq 0}} \varphi_k \psi_\ell \delta_{M^n k = \ell} = \sum_{k \neq 0} \varphi_k \psi_{M^n k}.$$

$$|\varphi_k \psi_{M^n k}| \leq |\varphi|_{C^1} |\psi|_{C^1} / M^{nk^2}.$$

$$\text{The error} \leq \sum_{|k| \neq 0} |\varphi|_{C^1} |\psi|_{C^1} / M^{nk^2} \leq M^{-n} |\varphi|_{C^1} |\psi|_{C^1} C.$$

Ques. What about higher dimensional case?

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Especially, when there are eigenvalue in the unit circle (Partially hyperbolic)

~~Gauss map: For $x \in [0, 1]$ let~~

~~$$f(x) = \frac{1}{x} - \left\lfloor \frac{1}{x} \right\rfloor$$~~

• Dynamical proof of mixing for cat map $M = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$

Let
$$v_1 = \begin{pmatrix} \frac{1+\sqrt{5}}{2} \\ 1 \end{pmatrix}, \quad v_2 = \begin{pmatrix} \frac{1-\sqrt{5}}{2} \\ 1 \end{pmatrix}$$

then $M v_i = \lambda_i v_i, \quad \lambda_1 = \frac{3+\sqrt{5}}{2}, \quad \lambda_2 = \frac{3-\sqrt{5}}{2}$

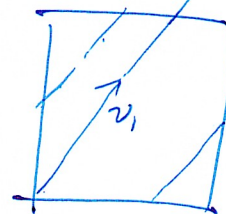
Therefore any $v \in \mathbb{R}^2$

$$M^n v = M^n (\alpha v_1 + \beta v_2) = \alpha \lambda_1^n v_1 + \beta \lambda_2^n v_2$$

So the map f_M preserves the foliation

given by v_i . (expanding / contracting foliations)

Let $h_i^s(x) = x + s v_i$, then



$$\star \quad f_M^n \circ h_i^s(x) = f_M^n(x) + s \lambda_i^n v_i = h_i^{s \lambda_i^n} \circ f_M^n(x)$$

Take φ, ψ C^1 functions on $\mathbb{T}^2$, then due to h_i^s preserving vol.

$$I_n = \int \varphi \psi \circ f_M^n = \frac{1}{t} \int_0^t \varphi(h_i^s x) \psi \circ f_M^n(h_i^s x) ds d\text{Leb}(x)$$

$$\approx \int \varphi(x) \left(\frac{1}{t} \int_0^t \psi \circ f_M^n \circ h_i^s(x) ds \right) d\text{Leb}(x) + |t| \cdot \|\varphi\|_C \|\psi\|_\infty$$

Now
$$\frac{1}{t} \int_0^t \psi \circ f_M^n \circ h_i^s(x) ds = \frac{1}{t} \int_0^t \psi \lambda_i^{ns} f_M^n(x) ds$$

Hence $\int \varphi(x) \frac{1}{t} \int_0^t \varphi(h_1^{\lambda_1^n s} f_m^n(x)) ds d\text{Leb}(x)$

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$$= \int \varphi(f_m^{-n}(x)) \cdot \frac{1}{t} \int_0^t \varphi(h_1^{\lambda_1^n s}(x)) ds d\text{Leb}(x)$$

By the unique ergodic of $h_1^1 \Rightarrow$ flow unique ergodic

$$\frac{1}{t} \int_0^t \varphi(h_1^{\lambda_1^n s}(x)) ds = \frac{1}{t\lambda_1^n} \int_0^{t\lambda_1^n} \varphi(h_1^s(x)) ds$$

$$\longrightarrow \int \varphi d\text{Leb}$$

Therefore

$$I_n \xrightarrow{n \rightarrow \infty} \int \varphi \int \varphi$$

Rk: Dynamical proof, use property of flow h_1^s and relation (*)

Similar idea also appears in unipotent flow and geodesic flow

• Gauss Map:

For $x \in (0, 1]$ let $a_1(x) = \left[\frac{1}{x} \right]$, Gauss map $f(x) = \frac{1}{x} - \left[\frac{1}{x} \right]$

$$a_n(x) = a_1(f^{n-1}(x))$$

By this way, we obtain continued fraction expansion of x

$$x = \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \frac{1}{a_4 + \dots}}}} = \frac{p_n + p_{n-1}x_n}{q_n + q_{n-1}x_n}$$

with $x_n = f^n(x)$

Due to $x_n = \frac{1}{a_{n+1} + x_{n+1}}$

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$$\frac{p_{n+1} + p_n x_{n+1}}{q_{n+1} + q_n x_{n+1}} = \frac{p_n (a_{n+1} + x_{n+1}) + p_{n-1}}{q_n (a_{n+1} + x_{n+1}) + q_{n-1}} = \frac{p_{n-1} + p_n a_{n+1} + p_n x_{n+1}}{q_{n-1} + q_n a_{n+1} + q_n x_{n+1}}$$

(*)
$$\begin{pmatrix} p_{n+1} & p_n \\ q_{n+1} & q_n \end{pmatrix} \rho^1 = \begin{pmatrix} p_n & p_{n-1} \\ q_n & q_{n-1} \end{pmatrix} \begin{pmatrix} a_{n+1} & 1 \\ 1 & 0 \end{pmatrix}$$

$$\begin{pmatrix} p_1 & p_0 \\ q_1 & q_0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ a_1 & 1 \end{pmatrix}.$$

lem. i) Continued fraction is finite iff $x \in \mathbb{Q}$

ii) the map $x \mapsto (a_1(x), a_2(x), \dots)$ is a bijection from $[0, 1] \setminus \mathbb{Q} \rightarrow (\mathbb{N} \setminus \{0\})^{\mathbb{N} \setminus \{0\}}$

i) ✓

Prf. ii) For $a = (a_1, \dots, a_n)$, let $\Delta_a = \{x \mid a_1(x), \dots, a_n(x) = a\}$

Then $f^n(\Delta_n) = [0, 1]$ and the inverse is given by

$$\psi_{\Delta_n} = f^{-n}|_{\Delta_n}(x) = \frac{p_n + p_{n-1}(x)}{q_n + q_{n-1}(x)} \quad (p, q \text{ given by } (*))$$

Hence $\Delta_n = \left(\frac{p_n}{q_n}, \frac{p_n + p_{n-1}}{q_n + q_{n-1}} \right)$ or $\left(\frac{p_n + p_{n-1}}{q_n + q_{n-1}}, \frac{p_n}{q_n} \right)$

$$h'(x) = \frac{p_{n-1}q_n - p_n q_{n-1}}{(q_n + q_{n-1}x)^2} = \frac{(-1)^{n-1}}{(q_n + q_{n-1}x)^2}.$$

$$\text{length}(\Delta_n) = \left| \frac{p_n + p_{n-1}}{q_n + q_{n-1}} - \frac{p_n}{q_n} \right| = \frac{1}{q_n(q_n + q_{n-1})} \rightarrow 0$$

as $q_n \rightarrow \infty$.

□

Ergodicity of Gauss map

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Prop.: The measure $\mu = \left(\frac{1}{\ln 2}\right) \frac{1}{1+x} dx$ is ergodic for f .
 Gauss map

Coro.: For a.e. $x \in (0, 1]$, for $l \in \mathbb{N} - \{0\}$

Step 1:

$$\frac{1}{n} \# \{ 1 \leq k \leq n \mid a_k(x) = l \} \xrightarrow{n \rightarrow \infty} \ln_2 \left(1 + \frac{1}{l+1} \right)$$

Pf. Let $A_l = \{ x \mid a_l(x) = l \}$ then $\mu(A_l) = \left(\frac{1}{\ln 2}\right) \left(\ln \left(1 + \frac{1}{l} \right) - \ln \left(1 + \frac{1}{l+1} \right) \right)$

$$= \frac{1}{\ln 2} \ln \left(1 + \frac{1}{l(l+1)} \right)$$

By Birkhoff ergodic thm.

$$\frac{1}{n} \# \{ 1 \leq k \leq n \mid a_k(x) = l \} = \frac{1}{n} \# \{ 1 \leq k \leq n \mid \text{ad } f^{k-1}(x) \in A_l \}$$

$$\longrightarrow \mu(A_l). \quad \square$$

Step 1. μ is f -invariant.

For any $\varphi \in L^1$ we have

$$\begin{aligned} \int \varphi \circ f d\mu &= \int_{(0,1]} \varphi(f(x)) \frac{1}{\ln 2} \frac{1}{1+x} dx \\ &= \frac{1}{n} \int_{\left[\frac{1}{n+1}, \frac{1}{n}\right]} \varphi\left(\frac{1}{x} - n\right) \frac{1}{\ln 2} \frac{1}{1+x} dx \\ \text{change of variable} \quad &= \frac{1}{n} \int_{(0,1]} \varphi(y) \frac{1}{\ln 2} \frac{1}{1 + \frac{1}{y+n}} \frac{1}{(y+n)^2} dy \\ &= \frac{1}{n} \int_{(0,1]} \varphi(y) \frac{1}{\ln 2} \left(\frac{1}{y+n} - \frac{1}{y+n+1} \right) dy \\ &= \int_{(0,1]} \varphi(y) \frac{1}{\ln 2} \frac{1}{1+y} dy. \quad \square \end{aligned}$$

Step 2, Renyi condition.

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~~For any~~ $\exists c, c' > 0$, for any Borel set A
and $a = (a_1, \dots, a_n) \in \mathbb{N}^n$ we have

$$c \mu(\Delta_a) \mu(A) \leq \mu(\Delta_a \cap f^{-n}(A)) \leq c' \mu(\Delta_a) \mu(A)$$

pf: μ and Leb differ by a bounded density suffices to consider Leb
Notice the map

$$f^{-n}|_{\Delta_a} : \Delta_a \rightarrow (0, 1]$$

is a diffeo and

$$\psi_a(x) = f^{-n}|_{\Delta_a}(x) = \frac{p_n + p_{n-1}x}{q_n + q_{n-1}x} \quad \text{inverse branch}$$

$$\left| (f^{-n}|_{\Delta_a})'(x) \right| = \left| \frac{(-1)^{n-1}}{(q_n + q_{n-1}x)^2} \right| \in \left(\frac{1}{(q_n + q_{n-1})^2}, \frac{1}{q_n^2} \right)$$

$$\mu(\text{Leb}(\Delta_a)) = \left| \frac{p_n}{q_n} - \frac{p_n + p_{n-1}}{q_n + q_{n-1}} \right| = \frac{1}{q_n(q_n + q_{n-1})}$$

Due to $q_n > q_{n-1}$, $\exists c, c' > 0$.

$$c (f^{-n}|_{\Delta_a}(x))' \leq \text{Leb}(\Delta_a) \leq c' (f^{-n}|_{\Delta_a}(x))'$$

Hence

$$\begin{aligned} \text{Leb}(\Delta_a \cap f^{-n}(A)) &= \text{Leb}(\psi_a^{-1}(A)) = \int_A |\psi_a'(x)| dx \\ &\in \text{Leb}(A) \left(\min_{\Delta_a} \psi_a', \max_{\Delta_a} \psi_a' \right) \\ &\in \text{Leb}(A) \text{Leb}(\Delta_a) (c, c') \quad \square \end{aligned}$$

Let Y be a f -inv Borel set

Step 3. Let $\nu(A) = \mu(A \cap Y) - \frac{c}{2} \mu(A) \mu(Y)$
with c from (Renyi Condition)

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For $A = \Delta_a$ we have

$$\begin{aligned} \nu(\Delta_a) &= \mu(\Delta_a \cap Y) - \frac{c}{2} \mu(\Delta_a) \mu(Y) \\ &= \mu(\Delta_a \cap f^{-n}(Y)) - \frac{c}{2} \mu(\Delta_a) \mu(Y) \geq 0 \\ &\geq \frac{c}{2} \mu(\Delta_a) \mu(Y) \end{aligned}$$

(Δ_a) a Bool semi-algebra generates Borel σ -algebra

$\forall A$ Borel $\forall \varepsilon > 0$. $\exists A'$ finite union of Δ_a

$$\mu(A \Delta A') < \varepsilon$$

So $\forall A$ Borel

$$\nu(A) \geq 0.$$

Take $A = Y^c$. $\nu(Y^c) = -\frac{c}{2} \mu(Y^c) \mu(Y) \geq 0$

Hence $\mu(Y)$ or $\mu(Y^c) = 0$

□

• Thm Gauss map is mixing.

$$\exists c, c' > 0$$

lem: For any Borel sets A, B

$$c' \mu(A) \mu(B) \leq \liminf_n \mu(A \cap f^n B) \leq \limsup_n \mu(A \cap f^{-n}(B)) \leq c \mu(A) \mu(B)$$

pf: For $\Delta = \Delta_a$, $\forall m > n$ we can decompose

$$\Delta_a = \bigcup_{\text{length}(b) = m} \Delta_b,$$

By Renyi condition $c' \mu(A) \mu(B) \leq \mu(\Delta_a \cap f^{-m}(B)) \leq c \mu(A) \mu(B)$

Then by approximation of Borel sets

□

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Pf of mixing: Let $\nu = \mu \times \mu$ and $g = f \times f$.

Let $\nu_n = (id \times f^n)_* \mu$

Only need to prove for any weak limit ν_∞ of $\{\nu_n\}_n$
we have $\nu_\infty = \nu$.

Step 1: ν is g -ergodic

pf: For any cylinder set $P = A \times B$, $Q = A' \times B'$

$$\text{we have } \liminf \nu(P \times g^{-n}Q) \\ = \liminf \mu(A \times f^{-n}A') \mu(B \times f^{-n}B')$$

$$\geq \underline{c} \nu(P) \nu(Q).$$

Use cylinder sets to approximate Borel sets

$$\text{we have } \liminf \nu(P \times g^{-n}Q) \geq \underline{c} \nu(P) \nu(Q)$$

For any g -inv set Y .

$$0 = \liminf \nu(Y \times g^{-n}Y) \geq \underline{c} \nu(Y) \nu(Y').$$

□

Step 2: $\nu_\infty(A \times B) \leq \limsup \nu_n(A \times B)$

$$= \limsup \mu(A \times f^{-n}B) \leq \underline{c}' \mu(A) \mu(B) \\ = \underline{c}' \nu(A \times B)$$

$$\Rightarrow \forall P \text{ Borel } \nu_\infty(P) \leq \nu(P)$$

Hence ν_∞ is absolute continuous w.r.t ν .

$$d\nu_\infty(x) = \overset{p(x)}{d\nu(x)}$$

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for $p \in L^1(\nu)$. By Radon-Nikodym thm.

But $p \nu_\infty$ is g -inv. and ν is g -erg.

Hence $p = \text{cst}$ a.e. So $\nu_\infty = \nu$.

□